

Volume 05 - Engineering Standards

Version v24 draft

README.md

Volume 05 - Engineering Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-README

Purpose

Establish the official engineering standards for Mariam AI Enterprise OS.

Scope

Volume 05 defines engineering standards for architecture, documentation, naming, versioning, repositories, code, Python, runtime, APIs, events, storage, security, testing, quality, observability, PDF builds, releases, Git, CI/CD, and governance.

Mandatory Rules

- Engineering work must follow the Source of Truth hierarchy in this repository.
- Standards apply to documentation and future implementation work.
- Exceptions must be explicit, temporary, owned, and traceable.

Recommended Practices

- Apply standards during design, not only during final review.
- Keep evidence lightweight but complete.
- Use the canonical dictionary when choosing names.

Prohibited Practices

- Do not implement against undocumented assumptions.
- Do not bypass security, testing, or release evidence because a change appears small.
- Do not treat generated artifacts as authoritative when Markdown sources disagree.

Required Evidence

- Updated standards files.
- Generated PDF, ZIP, manifest, and release notes.
- Verification that every standard contains mandatory sections.

Acceptance Criteria

- Administrative files and all 20 standards exist.

- Each standard contains practical mandatory rules, recommended practices, prohibited practices, required evidence, acceptance criteria, traceability, and version history.
- v24 release artifacts are generated and verified.

Traceability

- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial Volume 05 administrative document.

Volume 05 Index

Version: v24 draft

Status: Draft

Document ID: MAL-V05-INDEX

Purpose

List the canonical standards files and document IDs.

Scope

Volume 05 defines engineering standards for architecture, documentation, naming, versioning, repositories, code, Python, runtime, APIs, events, storage, security, testing, quality, observability, PDF builds, releases, Git, CI/CD, and governance.

Mandatory Rules

- Engineering work must follow the Source of Truth hierarchy in this repository.
- Standards apply to documentation and future implementation work.
- Exceptions must be explicit, temporary, owned, and traceable.

Recommended Practices

- Apply standards during design, not only during final review.
- Keep evidence lightweight but complete.
- Use the canonical dictionary when choosing names.

Prohibited Practices

- Do not implement against undocumented assumptions.
- Do not bypass security, testing, or release evidence because a change appears small.
- Do not treat generated artifacts as authoritative when Markdown sources disagree.

Required Evidence

- Updated standards files.
- Generated PDF, ZIP, manifest, and release notes.
- Verification that every standard contains mandatory sections.

Acceptance Criteria

- Administrative files and all 20 standards exist.
- Each standard contains practical mandatory rules, recommended practices, prohibited practices, required evidence, acceptance criteria, traceability, and version history.
- v24 release artifacts are generated and verified.

Traceability

- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial Volume 05 administrative document.

Standards Files

- MAL-V05-STD-001 - standard-05-01-architecture-standards.md
- MAL-V05-STD-002 - standard-05-02-documentation-standards.md
- MAL-V05-STD-003 - standard-05-03-naming-standards.md
- MAL-V05-STD-004 - standard-05-04-versioning-standards.md
- MAL-V05-STD-005 - standard-05-05-repository-standards.md
- MAL-V05-STD-006 - standard-05-06-code-standards.md
- MAL-V05-STD-007 - standard-05-07-python-standards.md
- MAL-V05-STD-008 - standard-05-08-runtime-standards.md
- MAL-V05-STD-009 - standard-05-09-api-standards.md
- MAL-V05-STD-010 - standard-05-10-event-standards.md
- MAL-V05-STD-011 - standard-05-11-storage-standards.md
- MAL-V05-STD-012 - standard-05-12-security-standards.md
- MAL-V05-STD-013 - standard-05-13-testing-standards.md
- MAL-V05-STD-014 - standard-05-14-quality-standards.md
- MAL-V05-STD-015 - standard-05-15-observability-standards.md
- MAL-V05-STD-016 - standard-05-16-pdf-build-standards.md
- MAL-V05-STD-017 - standard-05-17-release-standards.md
- MAL-V05-STD-018 - standard-05-18-git-standards.md
- MAL-V05-STD-019 - standard-05-19-ci-cd-standards.md
- MAL-V05-STD-020 - standard-05-20-governance-standards.md

Volume 05 Changelog

Version: v24 draft

Status: Draft

Document ID: MAL-V05-CHANGELOG

Purpose

Record engineering standard additions, revisions, and releases.

Scope

Volume 05 defines engineering standards for architecture, documentation, naming, versioning, repositories, code, Python, runtime, APIs, events, storage, security, testing, quality, observability, PDF builds, releases, Git, CI/CD, and governance.

Mandatory Rules

- Engineering work must follow the Source of Truth hierarchy in this repository.
- Standards apply to documentation and future implementation work.
- Exceptions must be explicit, temporary, owned, and traceable.

Recommended Practices

- Apply standards during design, not only during final review.
- Keep evidence lightweight but complete.
- Use the canonical dictionary when choosing names.

Prohibited Practices

- Do not implement against undocumented assumptions.
- Do not bypass security, testing, or release evidence because a change appears small.
- Do not treat generated artifacts as authoritative when Markdown sources disagree.

Required Evidence

- Updated standards files.
- Generated PDF, ZIP, manifest, and release notes.
- Verification that every standard contains mandatory sections.

Acceptance Criteria

- Administrative files and all 20 standards exist.
- Each standard contains practical mandatory rules, recommended practices, prohibited practices, required evidence, acceptance criteria, traceability, and version history.
- v24 release artifacts are generated and verified.

Traceability

- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial Volume 05 administrative document.

Volume 05 Traceability

Version: v24 draft

Status: Draft

Document ID: MAL-V05-TRACEABILITY

Purpose

Map engineering standards to constitutional, dictionary, roadmap, and blueprint authority.

Scope

Volume 05 defines engineering standards for architecture, documentation, naming, versioning, repositories, code, Python, runtime, APIs, events, storage, security, testing, quality, observability, PDF builds, releases, Git, CI/CD, and governance.

Mandatory Rules

- Engineering work must follow the Source of Truth hierarchy in this repository.
- Standards apply to documentation and future implementation work.
- Exceptions must be explicit, temporary, owned, and traceable.

Recommended Practices

- Apply standards during design, not only during final review.
- Keep evidence lightweight but complete.
- Use the canonical dictionary when choosing names.

Prohibited Practices

- Do not implement against undocumented assumptions.
- Do not bypass security, testing, or release evidence because a change appears small.
- Do not treat generated artifacts as authoritative when Markdown sources disagree.

Required Evidence

- Updated standards files.
- Generated PDF, ZIP, manifest, and release notes.
- Verification that every standard contains mandatory sections.

Acceptance Criteria

- Administrative files and all 20 standards exist.
- Each standard contains practical mandatory rules, recommended practices, prohibited practices, required evidence, acceptance criteria, traceability, and version history.
- v24 release artifacts are generated and verified.

Traceability

- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial Volume 05 administrative document.

Volume 05 Acceptance

Version: v24 draft

Status: Draft

Document ID: MAL-V05-ACCEPTANCE

Purpose

Define acceptance rules for the v24 engineering standards baseline.

Scope

Volume 05 defines engineering standards for architecture, documentation, naming, versioning, repositories, code, Python, runtime, APIs, events, storage, security, testing, quality, observability, PDF builds, releases, Git, CI/CD, and governance.

Mandatory Rules

- Engineering work must follow the Source of Truth hierarchy in this repository.
- Standards apply to documentation and future implementation work.
- Exceptions must be explicit, temporary, owned, and traceable.

Recommended Practices

- Apply standards during design, not only during final review.
- Keep evidence lightweight but complete.
- Use the canonical dictionary when choosing names.

Prohibited Practices

- Do not implement against undocumented assumptions.
- Do not bypass security, testing, or release evidence because a change appears small.
- Do not treat generated artifacts as authoritative when Markdown sources disagree.

Required Evidence

- Updated standards files.
- Generated PDF, ZIP, manifest, and release notes.
- Verification that every standard contains mandatory sections.

Acceptance Criteria

- Administrative files and all 20 standards exist.
- Each standard contains practical mandatory rules, recommended practices, prohibited practices, required evidence, acceptance criteria, traceability, and version history.
- v24 release artifacts are generated and verified.

Traceability

- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial Volume 05 administrative document.

Architecture Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-001

Purpose

Define mandatory engineering expectations for architecture boundaries, system decomposition, dependency direction, and blueprint compliance in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches architecture boundaries, system decomposition, dependency direction, and blueprint compliance.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-001

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Architecture Standards.

Documentation Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-002

Purpose

Define mandatory engineering expectations for documentation structure, required sections, review quality, and release readiness in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches documentation structure, required sections, review quality, and release readiness.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-002

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Documentation Standards.

Naming Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-003

Purpose

Define mandatory engineering expectations for canonical naming for files, document IDs, modules, events, services, APIs, and storage objects in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches canonical naming for files, document IDs, modules, events, services, APIs, and storage objects.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-003

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Naming Standards.

Versioning Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-004

Purpose

Define mandatory engineering expectations for version labels, compatibility, deprecation, migration, and release lineage in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches version labels, compatibility, deprecation, migration, and release lineage.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-004

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Versioning Standards.

Repository Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-005

Purpose

Define mandatory engineering expectations for repository layout, ownership, generated artifacts, workspace hygiene, and source-of-truth boundaries in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches repository layout, ownership, generated artifacts, workspace hygiene, and source-of-truth boundaries.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-005

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Repository Standards.

Code Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-006

Purpose

Define mandatory engineering expectations for code organization, readability, reviewability, error handling, and maintainability in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches code organization, readability, reviewability, error handling, and maintainability.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-006

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Code Standards.

Python Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-007

Purpose

Define mandatory engineering expectations for Python project structure, typing, dependency usage, scripts, and runtime compatibility in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches Python project structure, typing, dependency usage, scripts, and runtime compatibility.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-007

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Python Standards.

Runtime Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-008

Purpose

Define mandatory engineering expectations for runtime lifecycle, isolation, scheduling, recovery, observability, and policy enforcement in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches runtime lifecycle, isolation, scheduling, recovery, observability, and policy enforcement.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-008

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Runtime Standards.

API Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-009

Purpose

Define mandatory engineering expectations for API contracts, schemas, errors, pagination, authentication, idempotency, and compatibility in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches API contracts, schemas, errors, pagination, authentication, idempotency, and compatibility.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-009

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for API Standards.

Event Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-010

Purpose

Define mandatory engineering expectations for event naming, envelopes, ordering assumptions, correlation, retention, and consumer contracts in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches event naming, envelopes, ordering assumptions, correlation, retention, and consumer contracts.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-010

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Event Standards.

Storage Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-011

Purpose

Define mandatory engineering expectations for logical storage, schemas, retention, backup, migration, and data classification in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches logical storage, schemas, retention, backup, migration, and data classification.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-011

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Storage Standards.

Security Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-012

Purpose

Define mandatory engineering expectations for least privilege, tenant isolation, secrets, protected actions, audit, and threat controls in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches least privilege, tenant isolation, secrets, protected actions, audit, and threat controls.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-012

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Security Standards.

Testing Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-013

Purpose

Define mandatory engineering expectations for unit, integration, system, acceptance, security, performance, and regression testing in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches unit, integration, system, acceptance, security, performance, and regression testing.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-013

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Testing Standards.

Quality Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-014

Purpose

Define mandatory engineering expectations for quality gates, defect handling, usability, reliability, and acceptance evidence in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches quality gates, defect handling, usability, reliability, and acceptance evidence.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-014

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Quality Standards.

Observability Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-015

Purpose

Define mandatory engineering expectations for metrics, logs, traces, alerts, dashboards, incident evidence, and redaction in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches metrics, logs, traces, alerts, dashboards, incident evidence, and redaction.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-015

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Observability Standards.

PDF Build Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-016

Purpose

Define mandatory engineering expectations for Markdown to PDF generation, page verification, ZIP packaging, and manifest evidence in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches Markdown to PDF generation, page verification, ZIP packaging, and manifest evidence.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-016

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for PDF Build Standards.

Release Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-017

Purpose

Define mandatory engineering expectations for release notes, manifests, artifact naming, checksums, and release verification in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches release notes, manifests, artifact naming, checksums, and release verification.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-017

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Release Standards.

Git Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-018

Purpose

Define mandatory engineering expectations for branch hygiene, commits, history, generated files, push rules, and review traceability in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches branch hygiene, commits, history, generated files, push rules, and review traceability.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-018

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Git Standards.

CI CD Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-019

Purpose

Define mandatory engineering expectations for automation pipelines, gates, logs, artifacts, secrets, and deployment checks in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches automation pipelines, gates, logs, artifacts, secrets, and deployment checks.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
- Artifact names, checksums, and release notes for versioned releases.

Acceptance Criteria

- The standard can be applied by an engineer without additional verbal instruction.
- Mandatory rules, prohibited practices, and required evidence are specific enough for review.
- The standard aligns with Volumes 00 through 04 and does not contradict the Source of Truth model.
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Traceability

- MAL-V00-B010
- MAL-V00-B011
- MAL-V00-B012
- MAL-V00-B014
- MAL-V04-README
- MAL-V05-STD-019

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for CI CD Standards.

Governance Standards

Version: v24 draft

Status: Draft

Document ID: MAL-V05-STD-020

Purpose

Define mandatory engineering expectations for decision authority, exceptions, approvals, policy gates, and audit-ready engineering evidence in Mariam AI Enterprise OS.

Scope

This standard applies to documentation, specifications, implementation plans, code reviews, generated artifacts, tests, operations, and release evidence whenever work touches decision authority, exceptions, approvals, policy gates, and audit-ready engineering evidence.

Mandatory Rules

- Every change must trace to an approved document ID, roadmap phase, blueprint, specification, issue, or release note.
- Interfaces, events, storage records, and security controls must be documented before implementation.
- Generated artifacts must be reproducible from repository content and listed in MANIFEST.json.
- Security-sensitive behavior must include least-privilege controls, audit evidence, and failure behavior.
- Acceptance criteria must be written before a change is declared complete.
- Breaking changes must include migration, compatibility, and deprecation notes.
- Reviews must verify naming, versioning, tests, observability, and governance impact.

Recommended Practices

- Prefer small, reversible changes with clear ownership and limited blast radius.
- Use canonical dictionary terms from Volume 04 in names, comments, APIs, events, and dashboards.
- Keep examples close to real operating workflows instead of abstract placeholders.
- Add concise diagrams or tables when they reduce ambiguity.
- Automate validation for repeated checks such as required sections, PDF generation, ZIP contents, and manifest checksums.
- Record tradeoffs when a standard is intentionally relaxed.

Prohibited Practices

- Do not introduce code, configuration, event names, APIs, or storage schemas that conflict with the architecture library.
- Do not store secrets, credentials, private keys, tokens, or sensitive operational data in documentation or release artifacts.
- Do not ship generated artifacts without verifying readability and ZIP inclusion.
- Do not use undocumented autonomous behavior for AI, agents, workflows, providers, plugins, or connectors.

- Do not silently change document IDs, artifact names, or public contracts.
- Do not merge changes that lack tests or explicit risk acceptance.

Required Evidence

- Relevant source document IDs and traceability references.
- Updated Markdown source files when behavior, terminology, or architecture changes.
- Passing build output for PDFs, ZIP package, and MANIFEST.json.
- Test results appropriate to risk: unit, integration, security, acceptance, performance, or release packaging.
- Review notes for exceptions, unresolved risks, and compatibility decisions.
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Acceptance Criteria

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- Release artifacts include this standard in Markdown, PDF, ZIP, and manifest outputs.

Traceability

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- MAL-V04-README
- MAL-V05-STD-020

Version History

Version	Date	Status	Notes
v24 draft	2026-06-29	Draft	Initial engineering standard for Governance Standards.